Non-negative spatial matrix factorization for multi-sample spatial transcriptomics data

Yi Wang, Kasper Hansen, Department of Biostatistics, Johns Hopkins University

Method

The count of gene j at spot i in sample m is assumed to follow a Poisson distribution

$$y_{ij}^m \sim Poi(\gamma_i^m \lambda_{ij}^m)$$

where γ_i^m is the library size of spot i in sample m, and λ_{ij}^m is the normalized gene expression of gene j at spot i in sample m. The normalized gene expression of gene j at spot i in sample m is assumed to be a weighted linear combination of L non-negative latent factors.

$$\lambda_{ij}^m = \sum_{l=1}^{L} w_{jl} e^{f_{il}^m}$$

where $e^{f_{il}^m}$ is the value of lth latent factor at spot i in sample m. w_{ij} is the weight of factor l for gene j in spot i, which is the same between samples. The *log* of latent factor l at spot i in sample m follows a gaussian process distribution

$$f_{il}^m = f_l^m(x_i) \sim GP\big(\mu_l^m(x_i), \quad k_l^m(x_i)\big)$$

where k_l^m is the kernal of the lth factor at sample m. μ_l^m is the mean of factor l at sample m.

Results

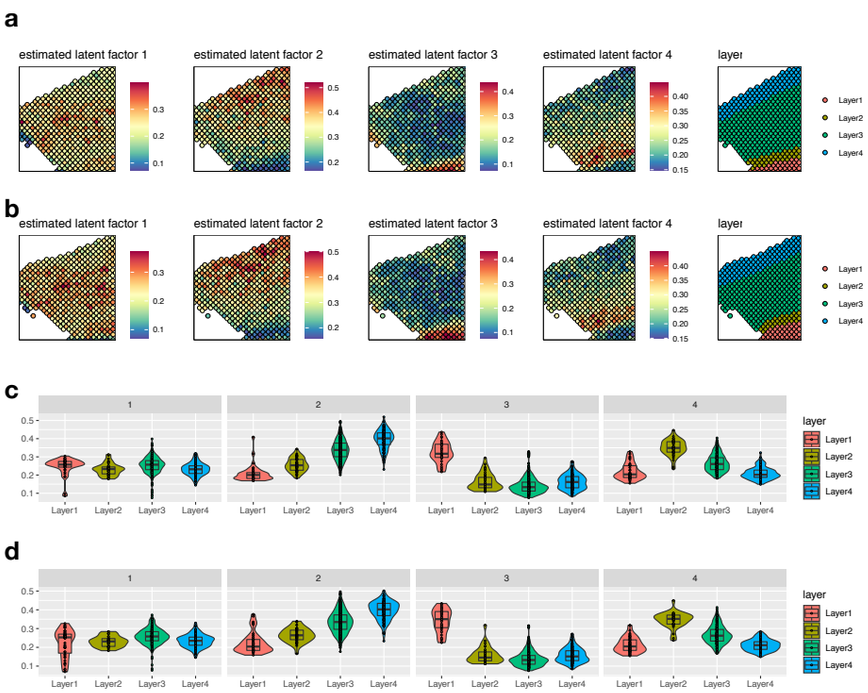

Fig 1. mNSF identifies shared layers | The spatial factors inferred by multi-sample non-negative spatial matrix factorization (mNSF) using two samples of spatial transcriptomics data with the same layer composition**.** Value of each spatial factor inferred by mNSF across spots in **a)** sample 1 and **b)** sample 2. Distribution of the value of each factor in the spots within each layer in **c)** sample 1 and **d)** sample 2.

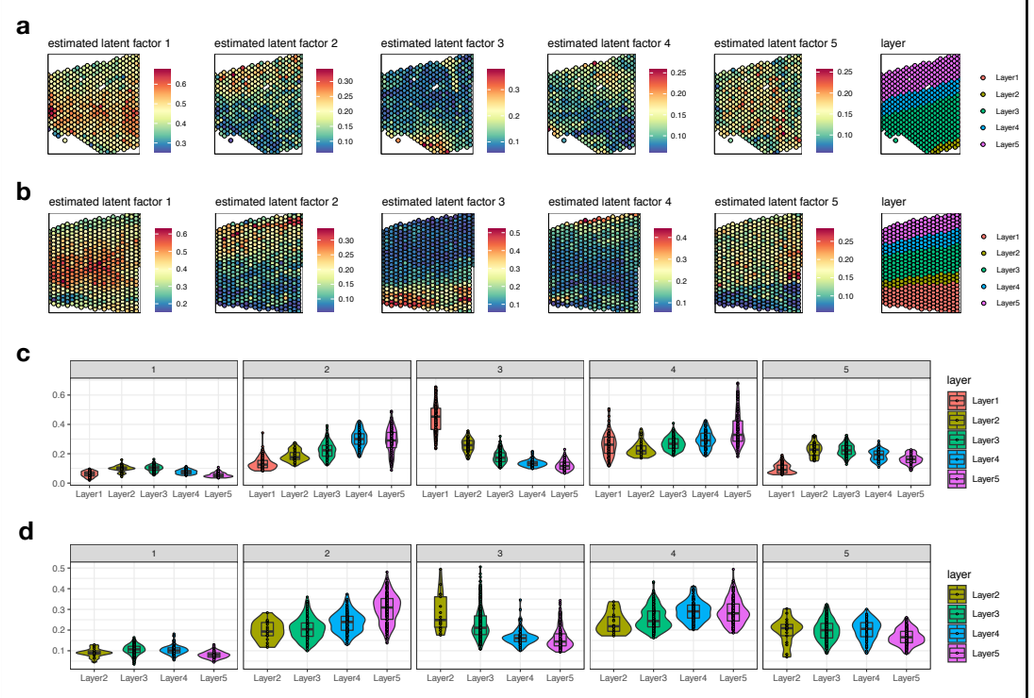

Fig 2. mNSF identifies sample-specific layers | The spatial factors inferred by mNSF using two samples of spatial transcriptomics data, with four layers shared by the two samples and one sample-specific layer. Value of each spatial factor, inferred by mNSF, across spots in **a)** sample 1 and **b)** sample 2. Distributions of the value of each factor in the spots within each layer in **c)** sample 1 and **d)** sample 2. The sample 2-specific layer is represented by the 3rd factor.

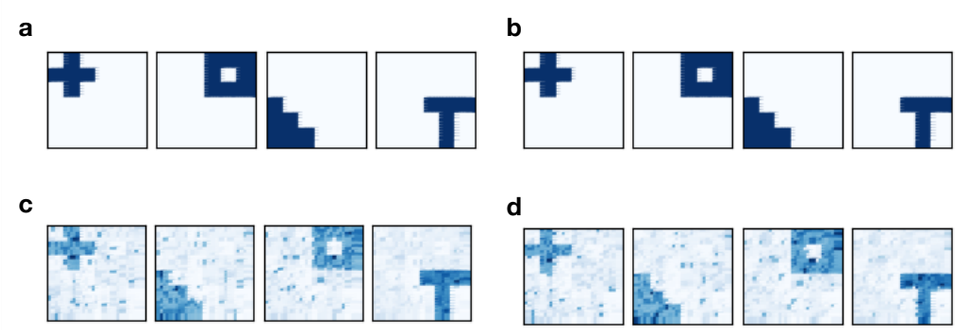

Fig 3. mNSF identifies shared factors between samples | The spatial factors inferred by mNSF using two samples of simulated data with shared spatial factors. True factors of **a)** sample 1 and **b)** sample 2 in the simulation settings. Value of each of the four spatial factors inferred by mNSF, across spots in **c)** sample 1 and **d)** sample 2.

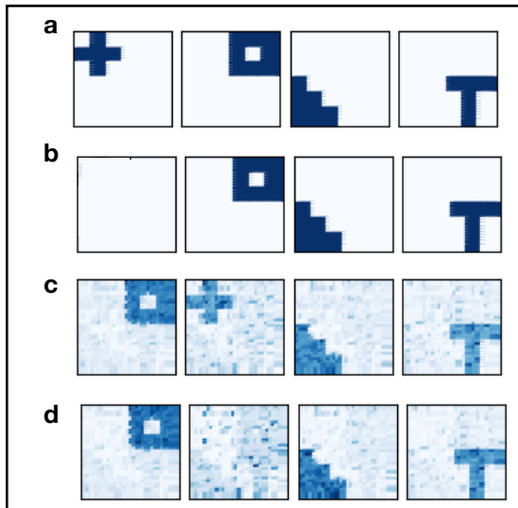

Fig 4. mNSF identifies sample-specific factors | The spatial factors inferred by mNSF using two samples of simulated data with three shared spatial factors and one sample-specific factor. True factors of **a)** sample 1 and **b)** sample 2 in the simulation settings. Value of each of the four spatial factors inferred by mNSF, across spots in **c)** sample 1 and **d)** sample 2.

Conclusions

We developed a method that extends non-negative spatial matrix factorization method (Townes, Engelhardt, 2021) to multi-sample spatial transcriptomics data. We applied our method in 2-sample spatial data as well as simulated data, and we show that our method identifies shared layers between samples and sample-specific layer when it exists.

Reference

William Townes, F., and Barbara E. Engelhardt. 2021. “Nonnegative Spatial Factorization.” *arXiv [stat.ME]*. arXiv. http://arxiv.org/abs/2110.06122.